

[START]

apsara

Identify TARGET TYPE

Date: . . .

TARGET = NUMERIC
(Regression Problem)

TARGET = Categorical
(Classification)

TARGET = Categorical
& Input Also CAT

Identify Input type

Identify Input type

Identify Input type
Classification

INPUT =
NUMERIC

INPUT =
CATEGORICAL

INPUT =
MIXED
(Num + Cat)

INPUT =
NUMERIC

INPUT =
CATEGORICAL

MODEL USED FOR CAT → CAT
CLASSIFICATION :

- Logistic Regression
- Decision TREE Class
- Random Forest Class
- KNN Class
- SVM Classifier

Preprocessing :
- No encoding needed
- Scaling optional

Preprocessing :
- Scale numeric feature
- Handle outliers

preprocessing for cat input
- One-Hot Encode cat
- Scale Numeric feature

Preprocessing :
- One-Hot Encoding I/P
- Label Encode target

USE :
Linear
Regression

USE :
Linear Reg +
One-Hot Encode
(CAT → NUM)

USE :
Linear Reg
or polynomial
Regression

USE :
Logistic
Regression
(NUM → CAT)
or :
- SVM
- KNN

USE :
Logistic Reg.
(CAT → CAT)
or :
- Decision Tree
- Random Forest

Final Step for Regression :
- Evaluate using RMSE, MAE, R^2
- Plot regression Line
- Check residuals

Evaluate Classification
- Accuracy
- Precision
- Recall
- F1 Score

1. Preprocessing / Feature Engineering

--> One-Hot Encoding

↳ Converts categorical values into binary vector

--> Label-Encoding

↳ Assign numeric labels to Categorical values.

--> ANOVA

Statistical test comparing means across groups.

--> ANOVA F-test

↳ Used for selecting important numeric features

--> Hypothesis Testing

↳ Evaluates assumption about data.

--> Qualitative Data

↳ Non-Numeric data (categorical)

--> Quantitative Data

↳ Numeric measurable data.

--> Exploratory Data analysis

↳ Understanding patterns in data.

--> Polynomial Features

↳ Create higher-degree terms for non-linear models.

--> Kernel tricks

↳ Transforms data into higher dimension (used in SVM)

2. SUPERVISED LEARNING

A. Regression Models (Predict numeric values)

↳ Linear Regression

↳ Models $[Y = mX + c]$

↳ Multiple Linear Regression

↳ Regression with multiple features

↳ Polynomial Regression

Fits non-linear curves using polynomial terms

↳ Ridge Regression

L2-regularized linear Regression

↳ Lasso Regression

L1-regularized linear Regression

↳ ElasticNet

Combination of L1 and L2 regularization

↳ Pickle File (.pkl)

Used to save trained ML models.

B. Classification Models (Predict class labels)

→ Logistic Regression

Binary and multi-class classification.

→ Naive Bayes

Probabilistic text classification.

→ KNN

Predicts based on closest neighbours.

→ SVM

Finds optimal separating hyperplane.

→ Decision Tree

Tree-based model using Gini/Entropy

→ Entropy

Impurity metric based on information gain.

→ Gini Index

Impurity metric measuring misclassification

→ Random Forest

Ensemble of many decision tree.

Evaluation Metrics

→ Confusion Matrix

True/False predictions visualization

→ Precision, Recall, F1-score

Quality of classification

→ Accuracy

Percentage of correct prediction.

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C. Ensemble Learning Methods

→ Bagging

Reduces variance by training multiple models.

→ Boosting

Sequential models that improve on previous errors.

→ ADABOOSTING

Weight-based boosting

→ Gradient Boosting

Builds trees sequentially using gradient

→ Catboost

Efficient boosting for categorical data.

→ Random Forest

Bagging of decision trees

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3. UNSUPERVISED LEARNING

A. Clustering Algorithms

→ k-means

Partitions data into k clusters.

→ DBSCAN

Density-based clustering, handles noise.

→ Heirarchical Clustering

Creates tree-like cluster structure.

→ Dendrogram

Visual representation of clustering merging.

4. Time Series Analysis

A. Smoothing Methods

→ Moving Average

Smooths data by averaging windows

→ Exponential Smoothing

Smooths data giving higher weight to recent points

B. Forecasting Components

→ Trend - Long-term movement in the data.

→ Seasonality - Repeating patterns.

→ Cycle Behaviour - Long-term cycles not fixed to a period.

5. GENERAL ML CONCEPTS

→ Overfitting

Model learns noise, performs poorly on test data.

→ Underfitting

Model is too simple, fails to learn patterns.

→ Feature Engineering

Creating/improving input features.

→ Random State

Ensures reproducible results.

→ Train/Test split

Separating data for training and evaluation.